# SDG Mapping to Project Components

## SDG 3: Good Health and Well-Being

**Project Contribution:**

- AI-based **multimodal diagnostic system**
- **Early disease detection**
- **Risk scoring & triage prioritization**
- **Transplant decision support**

**Mapping:**

| Project Feature | SDG Impact |
|---|---|
| CNN-based medical image analysis | Improves diagnostic accuracy |
| NLP-based lab report analysis | Enables faster clinical insights |
| Multimodal correlation engine | Reduces diagnostic errors |
| Transplant matching module | Improves patient survival outcomes |
| Fast inference (~4–5 sec) | Enables timely treatment decisions |

**Core Idea:** Enhances healthcare quality, speed, and patient safety.

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## SDG 9: Industry, Innovation, and Infrastructure

**Project Contribution:**

- Advanced **AI + NLP + CV integration**
- **Scalable architecture**
- **FHIR-compatible outputs**
- **Cloud-ready deployment**

**Mapping:**

| Project Feature | SDG Impact |
| --- | --- |
| Deep learning (ResNet, CNN) | Promotes technological innovation |
| NLP pipeline for clinical data | Advances healthcare AI systems |
| Modular layered architecture | Supports scalable infrastructure |
| Integration with EHR systems | Strengthens digital healthcare infrastructure |
| Cloud-deployable system | Enables modern, resilient systems |

**Core Idea:** Builds innovative and future-ready healthcare infrastructure.

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# SDG 10: Reduced Inequalities

**Project Contribution:**

- Reduces dependency on **specialists (radiologists)**
- Provides **AI-assisted diagnostics in low-resource areas**
- Standardizes clinical interpretation

**Mapping:**

| Project Feature | SDG Impact |
| --- | --- |
| Automated diagnosis support | Bridges specialist shortage |
| Structured clinical outputs | Ensures uniform healthcare quality |
| AI triage system | Improves access in rural/remote regions |
| Decision support system | Reduces geographic healthcare gaps |

**Core Idea:** Makes quality healthcare more accessible and equitable.

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## SDG 17: Partnerships for the Goals

**Project Contribution:**

- Requires collaboration between:
  - Healthcare providers
  - AI engineers
  - Policy makers
- Designed for **interoperability across systems**

**Mapping:**

| Project Feature | SDG Impact |
|---|---|
| Interoperable system (FHIR standards) | Enables cross-institution integration |
| Multimodal data integration | Encourages multi-domain collaboration |
| Deployment in hospital systems | Supports institutional partnerships |
| Data-driven healthcare ecosystem | Promotes shared innovation |

**Core Idea:** Encourages collaboration across sectors for healthcare advancement.

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# Final Summary (Concise Mapping)

| SDG | Theme | How Your Project Contributes |
|---|---|---|
| **SDG 3** | Health | Better diagnosis, faster triage, safer transplants |
| **SDG 9** | Innovation | AI-powered, scalable healthcare system |
| **SDG 10** | Equality | Accessible diagnostics in underserved areas |
| **SDG 17** | Partnership | Interoperable, collaborative healthcare ecosystem |